S.Y. B. Tech. Academic Year 2026-27 Semester: III

CSE20350/AID20210/CLD 20020: Data Structures Laboratory

WRITE UP

Experiment Number: 08

TITLE: Linear Queue

PROBLEM STATEMENT:

In a networked environment, a shared printer receives multiple print requests from users throughout the day. To handle these requests efficiently and ensure optimal memory usage, the
system implements a linear queue to manage the list of print jobs. Each print job is identified
by a Job ID and a Document Title. The print jobs are processed in the order they are received
(FIFO order), and memory is reused as jobs are completed

OBJECTIVE:

1. Implement a Linear Queue using an array.
2. Perform Enqueue, Dequeue and Display operations.
3. Understand FIFO behavior and overflow/underflow conditions

THEORY:

A queue is a linear data structure that follows the FIFO (First In, First Out) principle. This means the element inserted first is the first one to be removed, similar to a queue of people waiting for a bus or at a ticket counter.

In a queue:

- Insertion is performed at the Rear end.
- Deletion is performed at the Front end.

Types of Queue

1. Simple Queue
    - Follows FIFO.
    - Insertion at rear.
    - Deletion at front.
2. Circular Queue
    - Last position connects to the first position.
    - Efficiently utilizes array space by reusing freed positions.
3. Priority Queue
    - Elements are removed based on priority rather than insertion order.
    - Higher-priority elements are served first.
4. Double-Ended Queue (Deque)
    - Insertion and deletion are allowed at both the front and rear.
    - Types:
        - Input-Restricted Deque (insertion at one end only)
        - Output-Restricted Deque (deletion at one end only)

Applications of Queue

1. CPU process scheduling.
2. Printer spooling.
3. Breadth-First Search (BFS) in graphs.
4. Resource sharing.
5. Network packet buffering.
6. Customer service systems.
7. Interrupt handling in operating systems.

8. Simulation of real-world waiting lines.

IMPLEMENTATION:

- **PLATFORM:**
  - 64-bit Open source Linux or its derivatives.
  - Open Source C Programming tool like gcc/Eclipse Editor.

- **TEST CONDITIONS:-**

Test Case	Input	Expected Output
TC01	Create an empty queue	Queue initialized successfully.
TC02	Enqueue Job ID: 101, Report.pdf	Print job added successfully.
TC03	Enqueue print jobs until queue is full	All jobs inserted successfully until the queue reaches maximum capacity.
TC04	Enqueue one more job when queue is full	Queue Overflow! Print queue is full.
TC05	Display queue	All pending print jobs are displayed in FIFO order.
TC06	Dequeue one print job	The first inserted print job is processed and removed.
TC07	Dequeue all print jobs	All jobs are processed one by one until the queue becomes empty.
TC08	Dequeue from an empty queue	Queue Underflow! No print jobs available.
TC09	Display an empty queue	No Pending Print Jobs is displayed.
TC10	Exit the program	Program terminates successfully.

- **PSEUDO CODE:** ***//To be Written by Students***

  Write pseudocode for initialization, insertion (enqueue), deletion (dequeue), display, overflow checking, and underflow checking using a Linear Queue.

- **TIME COMPLEXITY:** ***//To be Written by Students***

  Write time complexity of above operations

- **CONCLUSION:**

The Printer Queue Manager using a Linear Queue successfully implements FIFO scheduling for print jobs. It performs Enqueue, Dequeue, and Display operations correctly while handling overflow and underflow conditions, demonstrating the practical application of the Linear Queue data structure in managing print requests.

FAQs *//To be Written by Students*

1. What is a Linear Queue?
2. What principle does a Linear Queue follow?
3. What are the basic operations performed on a Linear Queue?
4. What is Enqueue in a Linear Queue?
5. What is Dequeue in a Linear Queue?
6. How does a Linear Queue reuse memory?
7. What are the applications of a Linear Queue?
8. What is the time complexity of Enqueue and Dequeue operations?
9. What is the limitation of a Linear Queue?

Practice Assignments – Linear Queue

1 **Bank Customer Queue Management**
Develop a Linear Queue program to manage customers waiting at a bank counter. Implement Enqueue, Dequeue, and Display operations and handle queue overflow and underflow conditions.

2 **Supermarket Billing Queue**
Develop a Linear Queue program to manage customers waiting at a supermarket billing counter. Display the queue of customers and process them in FIFO order.

3 **Airport Check-in Queue**
Develop a Linear Queue application to manage passengers waiting for check-in. Display the current queue and process passengers using the FIFO principle.

4 **Document Scanning Queue**
Create a Linear Queue to manage documents waiting to be scanned. Allow users to add documents, process the next document, and display all pending documents.

5 **Online Course Registration Queue**
Develop a Linear Queue program to manage student registration requests for

an online course. Implement Enqueue, Dequeue, and Display operations and handle overflow and underflow conditions.